2C Irrational Numbers and the Real Number System

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Classify each real number as Rational ($\mathbb{Q}$) or Irrational ($\mathbb{I}$):

(a) $\sqrt{121}$ (b) $\sqrt{40}$ (c) $-0.\dot{8}\dot{1}$ (d) 3π

Q2. Evaluate each exact root:

(a) $\sqrt{169}$ (b) $\sqrt{\dfrac{49}{144}}$ (c) $\sqrt[3]{-125}$ (d) $\sqrt{0.36}$

Q3. Find the two consecutive integers between which each irrational root lies:

(a) $\sqrt{31}$ (b) $\sqrt{88}$ (c) $-\sqrt{55}$ (d) $\sqrt[3]{50}$

Q4. Evaluate each expression involving exact square and cube roots:

(a) $\sqrt{144} - \sqrt[3]{-64}$ (b) $(\sqrt{19})^2 + \sqrt{81}$ (c) $\sqrt{9^2 + 12^2}$ (d) $2\sqrt{225} \div \sqrt[3]{27}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A square courtyard in Fitzroy has an area of $75\ \text{m}^2$. (a) Write the exact side length of the courtyard as a surd. (b) Between which two consecutive whole metres does the side length lie?

Q6. A circular picnic lawn at the Royal Botanic Gardens Victoria has radius $r = 11\ \text{m}$. Write its exact circumference $C = 2\pi r$ and exact area $A = \pi r^2$ in terms of π, and state whether these exact values are rational or irrational.

Q7. Compare the common rational approximation $\dfrac{22}{7}$ with $\pi \approx 3.14159265\ldots$ to 4 decimal places. Which is larger, $\dfrac{22}{7}$ or π?

Q8. A cubic shipping crate has a volume of $343\ \text{L}$ ($0.343\ \text{m}^3$). Find the exact internal side length of the crate in centimetres ($1\ \text{L} = 1000\ \text{cm}^3$).

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. Give an example of two irrational numbers whose product is a rational number, and an example of two irrational numbers whose sum is a rational number.

Q10. Explain why $\sqrt{-36}$ is not a real number, whereas $\sqrt[3]{-64} = -4$ IS a real number.